

TRAVEL & TRANSPORTATION (AIRLINES)

BIG DATA ANALYTICS IN AVIATION INDUSTRY

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MBA(GENERAL)

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INTRODUCTION & PROBLEM STATEMENT

Problem Statement:

- Airlines face challenges in flight delays, customer dissatisfaction, route inefficiencies, and volatile pricing.
- Growing data from operations, customers, and social media is underutilized.

Project Objective:

- Use Big Data Analytics to identify actionable insights that can enhance: Operational efficiency, Customer experience, Network optimization, Brand sentiment, Pricing strategies

Tools & Technologies:

- Python (Pandas, Plotly), Databricks, Excel, PySpark



METHODOLOGY AND DATASET OVERVIEW

Methodology Steps:

- 1. Data Collection:** Collected 5 real aviation datasets from Kaggle and Open Data Portals.
- 2. Data Cleaning & Preparation:** Removed nulls and duplicates at scale. Renamed inconsistent columns for schema alignment. Converted categorical and timestamp fields for analysis.
- 3. Exploratory Data Analysis (EDA):** Used PySpark + Pandas for large - scale data aggregation. Visualized patterns through Plotly
- 4. Network Analysis:** Applied GraphFrames to model global flight routes. Identified key hubs using degree centrality and PageRank.
- 5. Dashboard Creation & Insights:** Consolidate all 5 use cases integrating outputs from Spark -based analysis into a single visualization interface.

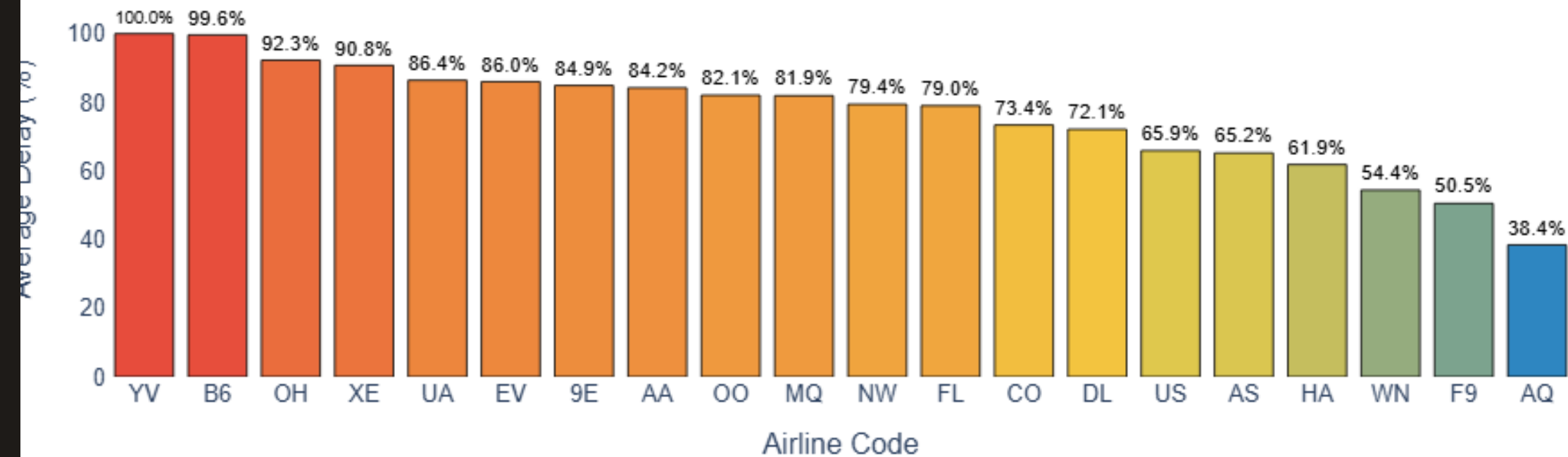
Datasets Used:

- 1. Airlines Delay** - Operational Efficiency
- 2. Passenger Satisfaction** - Customer Experience
- 3. Flight Route Database** - Route Optimization
- 4. Twitter US Airline Sentiment** - Brand Perception
- 5. Flight Price Prediction** - Revenue Management

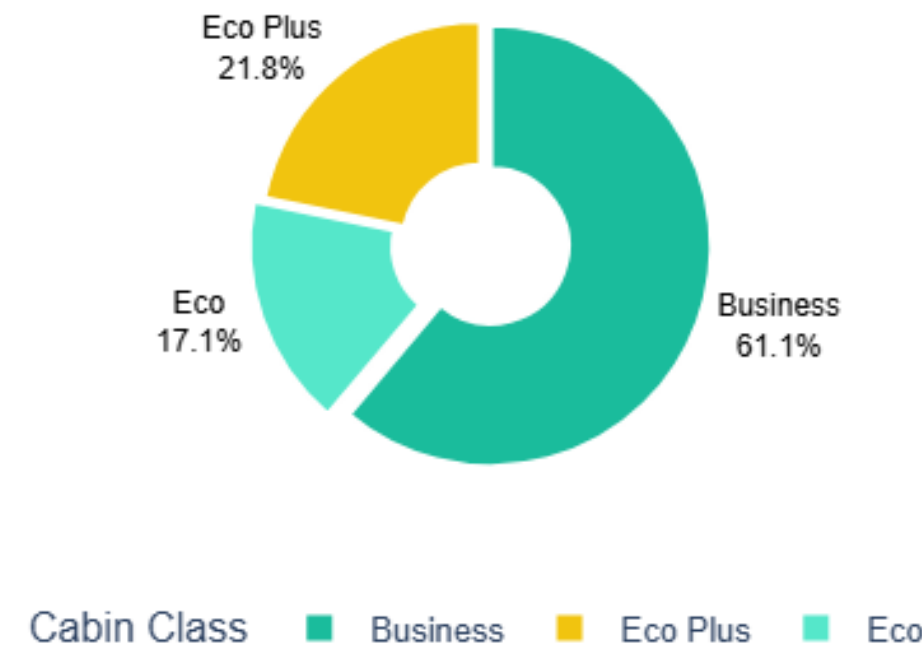


ANALYTICAL INSIGHTS – PART 1

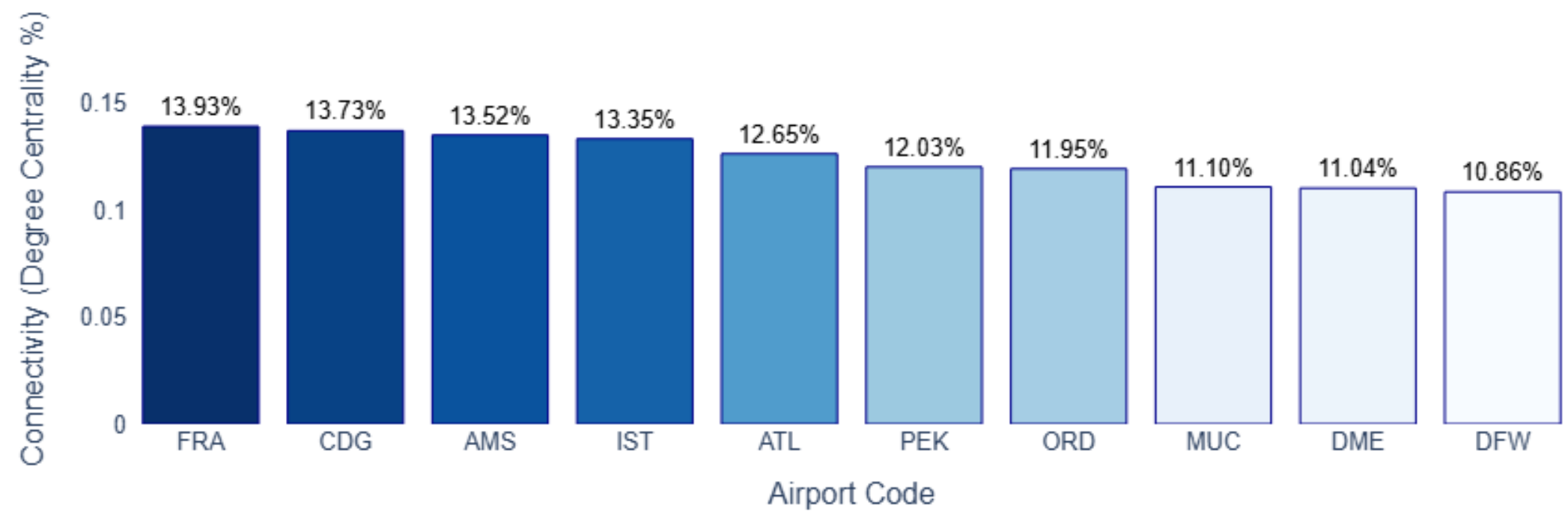
✈ Case 1: Average Flight Delay by Airline



😊 Case 2: Passenger Satisfaction by Class



🌐 Case 3: Top 10 Most Connected Airports (Network Centrality)



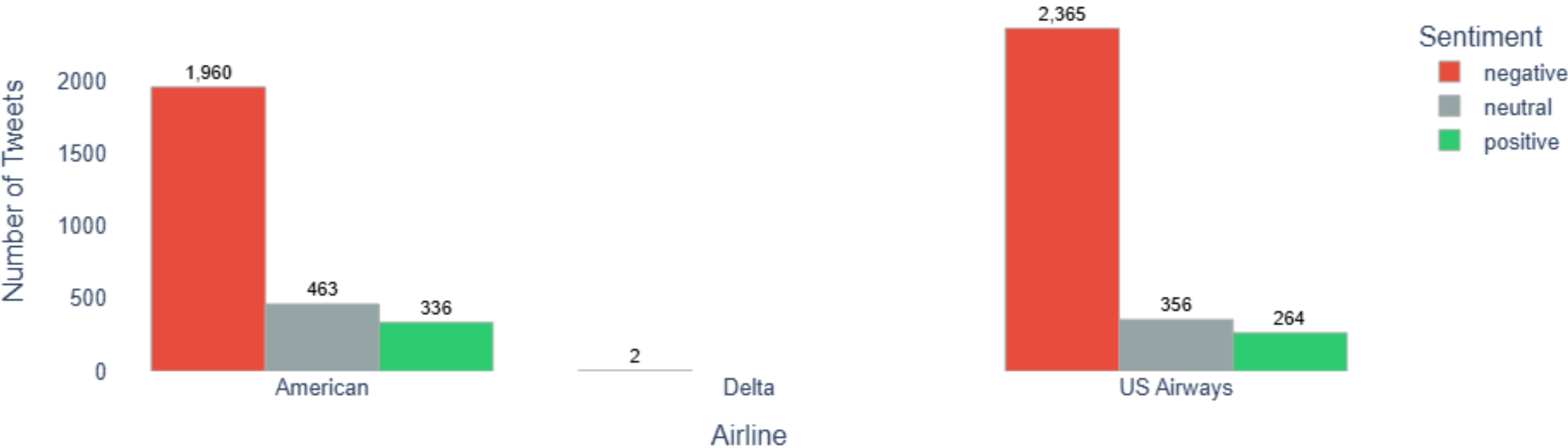
FINDING 1: AIRLINES WITH HIGHER DELAY %→ NEED PREDICTIVE MAINTENANCE AND SCHEDULING.

FINDING 2: BUSINESS CLASS PASSENGERS MOST SATISFIED.

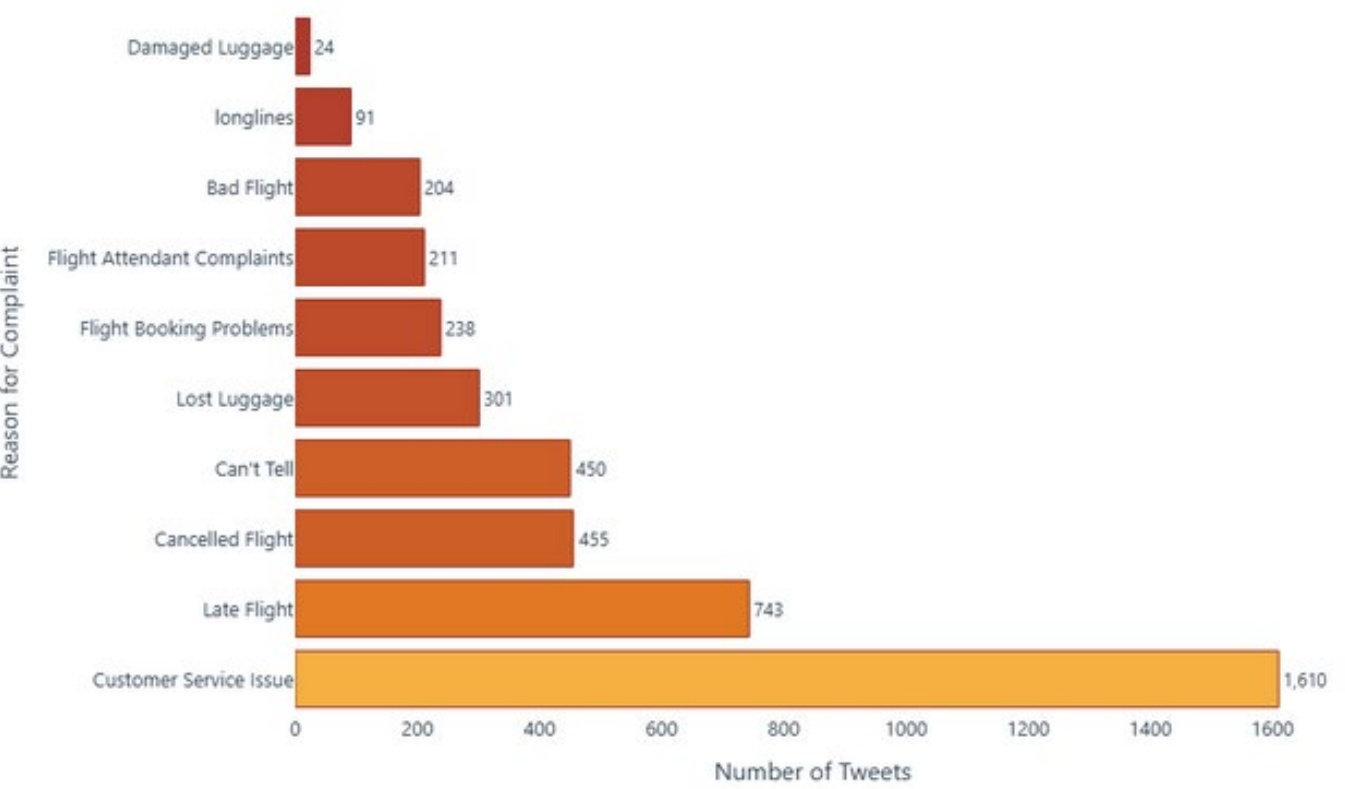
FINDING 3: LHR, ATL, DXB ARE TOP HUBS FOR GLOBAL CONNECTIVITY.

ANALYTICAL INSIGHTS – PART 2

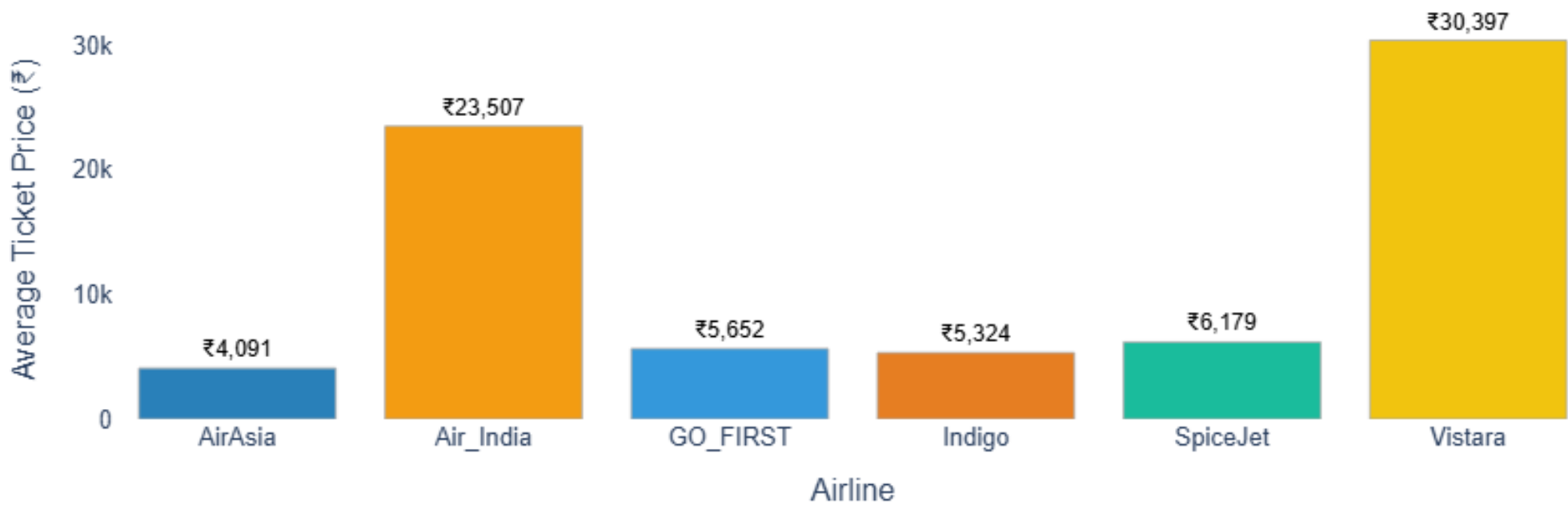
Case 4: Airline-wise Sentiment Breakdown



Case 4: Top 10 Reasons for Negative Airline Tweets



Case 5: Average Flight Price by Airline



FINDING 4: MOST COMPLAINTS RELATE TO DELAYS AND CUSTOMER SERVICE.
BUSINESS USE: AI-BASED FEEDBACK MONITORING CAN ENHANCE BRAND TRUST.

FINDING 5: BUSINESS FARES ARE 3× HIGHER THAN ECONOMY; SOME AIRLINES SHOW STEEP VARIATION.
BUSINESS USE: DYNAMIC PRICING CAN MAXIMIZE PROFIT AND DEMAND BALANCE.

KEY TAKEAWAYS, RECOMMENDATIONS & CONCLUSION

Key Takeaways

- **PySpark Analysis** : Helped identify major flight delays, customer satisfaction trends, and pricing patterns efficiently .
- **Passenger Insights** : Business Class shows highest satisfaction consistent service drives loyalty .
- **Network Findings** : Key airports like LHR, ATL, and DXB are major global hubs for route optimization .
- **Social Sentiment** : Most complaints relate to delays sentiment analytics can improve brand image .
- **Dynamic Pricing** : Business fares $\approx 3 \times$ Economy real-time data can optimize fare strategies .

Recommendations

- Use predictive models in PySpark MLlib to reduce delays.
- Apply real-time sentiment monitoring for quick service recovery.
- Implement dynamic pricing using Spark streaming data.
- Focus on hub efficiency and route planning for better utilization.

Conclusion

Big Data with PySpark enables faster, smarter airline decisions improving punctuality, satisfaction, and profitability.

By integrating flight, passenger, route, and social data, airlines can truly fly smarter and serve better.



THANK
YOU

